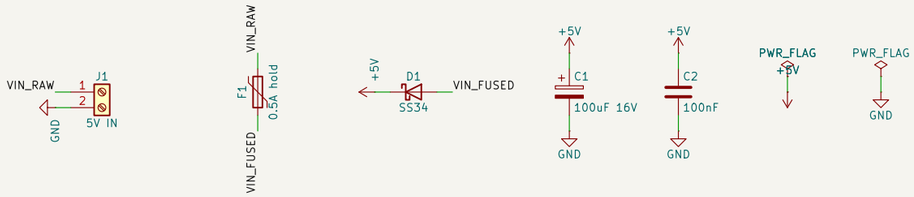
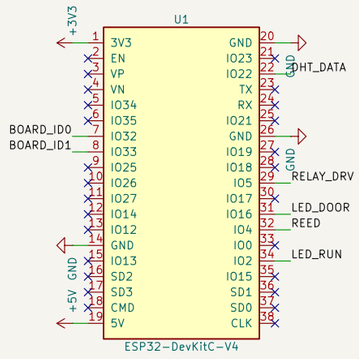
POWER: 5 V in -> PTC fuse -> Schottky (reverse-polarity + USB back-feed block) -> +5V

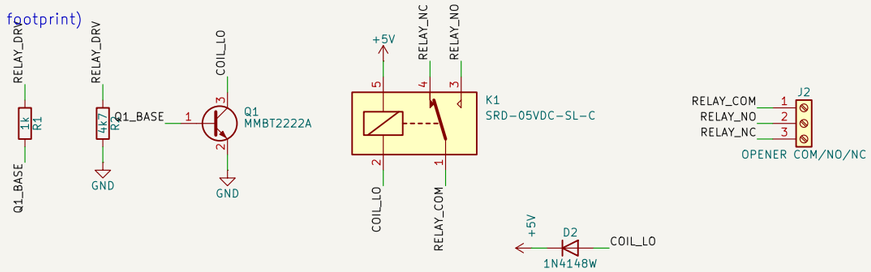


NOTES
1. Firmware unchanged: GPIO map = main/board.h. Reset/factory button = devkit Boot button (GPIO0).
2. External 5 V (J1) and USB together only if the devkit has a diode on USB VBUS - check first.
3. Devkit rows are 25.4 mm apart (DevKitC V4) - measure the real board before ordering PCBs.
4. Relay output is for the opener's low-voltage push-button input. Not for mains.
5. Bring-up checklist: docs/superpowers/specs/2026-10-16-carrier-pcb-schematic-design.md

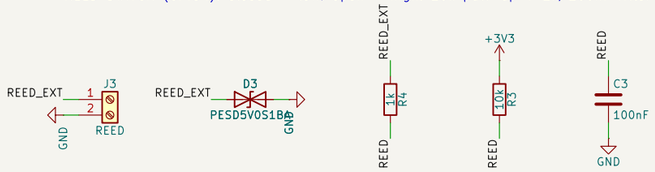
RELAY (GPIO5): R2 4k7 pull-down holds GPIO5 ~0.3 V during boot (internal ~45k pull-up), below Q1 Vbe -> relay cannot pull in while the ESP32 resets. J2: wire COM+NO across the opener's wall button.

ESP32 DevKitC V4 (WROOM-32U) - plugs into 2x 1x19 sockets (U1 footprint)

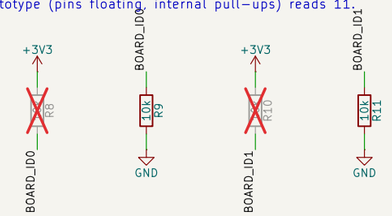




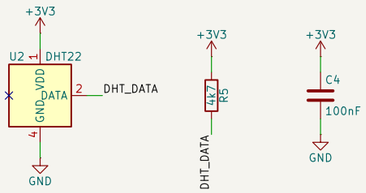
REED SWITCH (GPIO4): closed = low, open = high. 10k pull-up + 1k/100nF filter (~1 ms) + ESD at J3.



BOARD ID (GPIO32 = ID0, GPIO33 = ID1), not read by firmware yet.
Fit exactly one resistor per line. Rev 1 = 00: R9, R11 fitted; R8, R10 DNP.
Prototype (pins floating, internal pull-ups) reads 11.



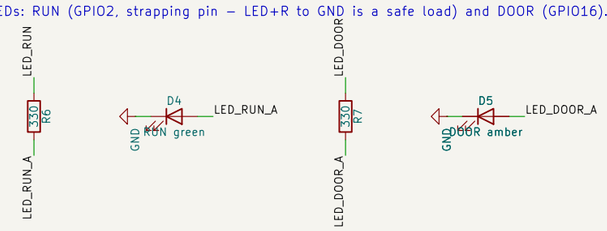
DHT22 / AM2302 (GPIO22), on board. Place at the far end from U1/K1 on a slotted neck; vent the enclosure.



MOUNTING: 4x M3



LEDs: RUN (GPIO2, strapping pin - LED+R to GND is a safe load) and DOOR (GPIO16). ~4 mA each.



Lars-Ake Ekstrand
ESP32 DevKitC V4 (WROOM-32U) carrier - GPIO map = main/board.h

Sheet: /
File: garage-opener-pcb.kicad_sch

Title: Garage opener carrier PCB

Size: A3 | Date: 2026-10-16 | **Rev: 1**
KiCad E.D.A. 10.0.6 | Id: 1/1